# A Simulated Annealing Algorithm for Maximum Common Edge Subgraph Detection in Biological Networks

GECCO 2016

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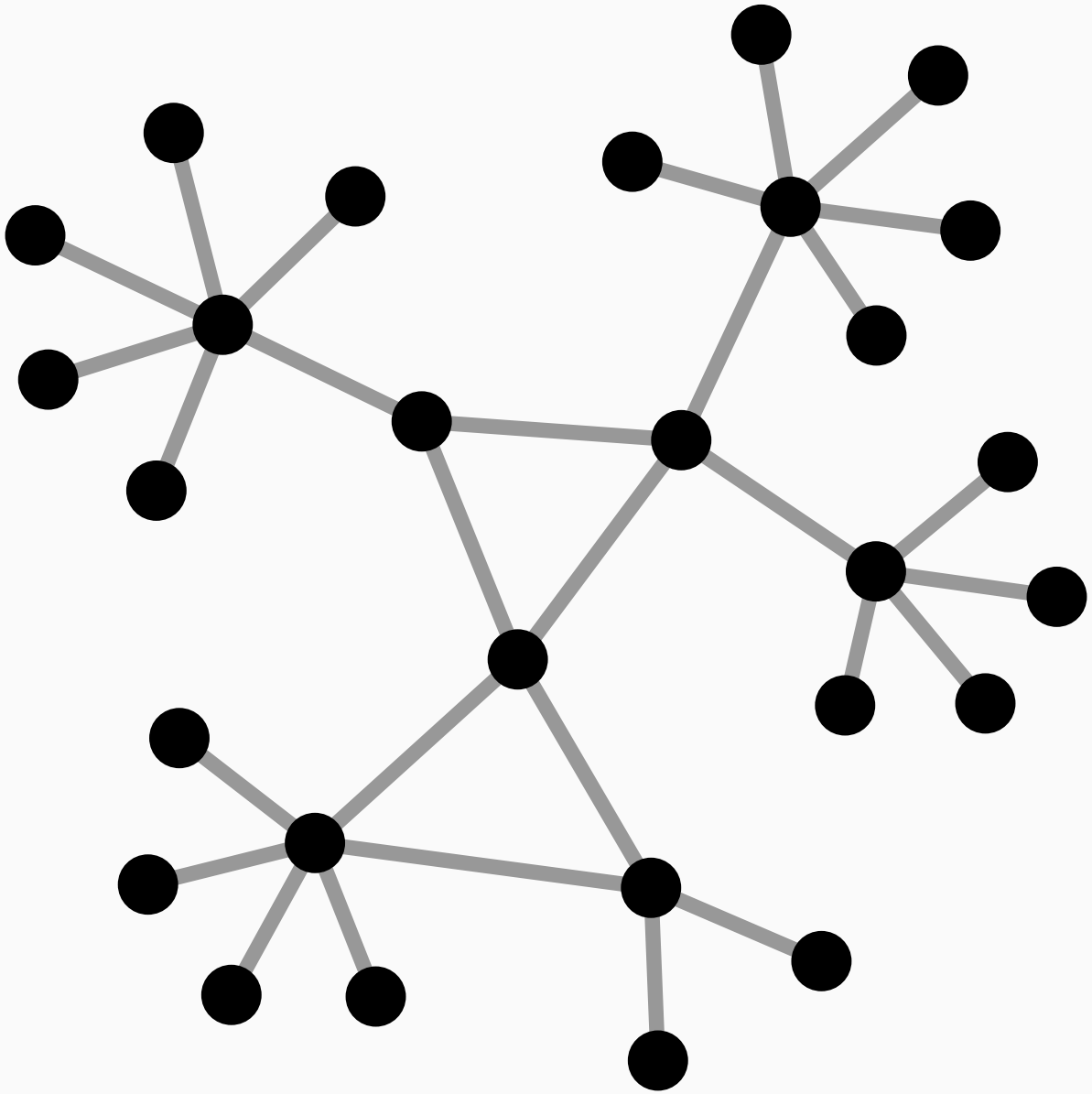
# Networks



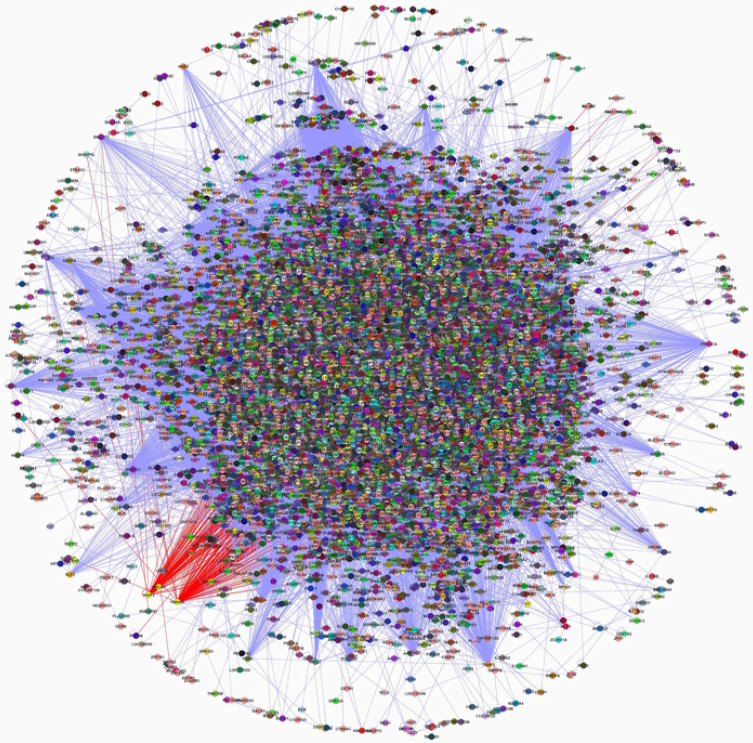
# Background

Goal: Detect a large common subgraph between multiple graphs.

A fundamental graph theoretical problem. MCES is *NP-hard*.

Focus on biological networks:

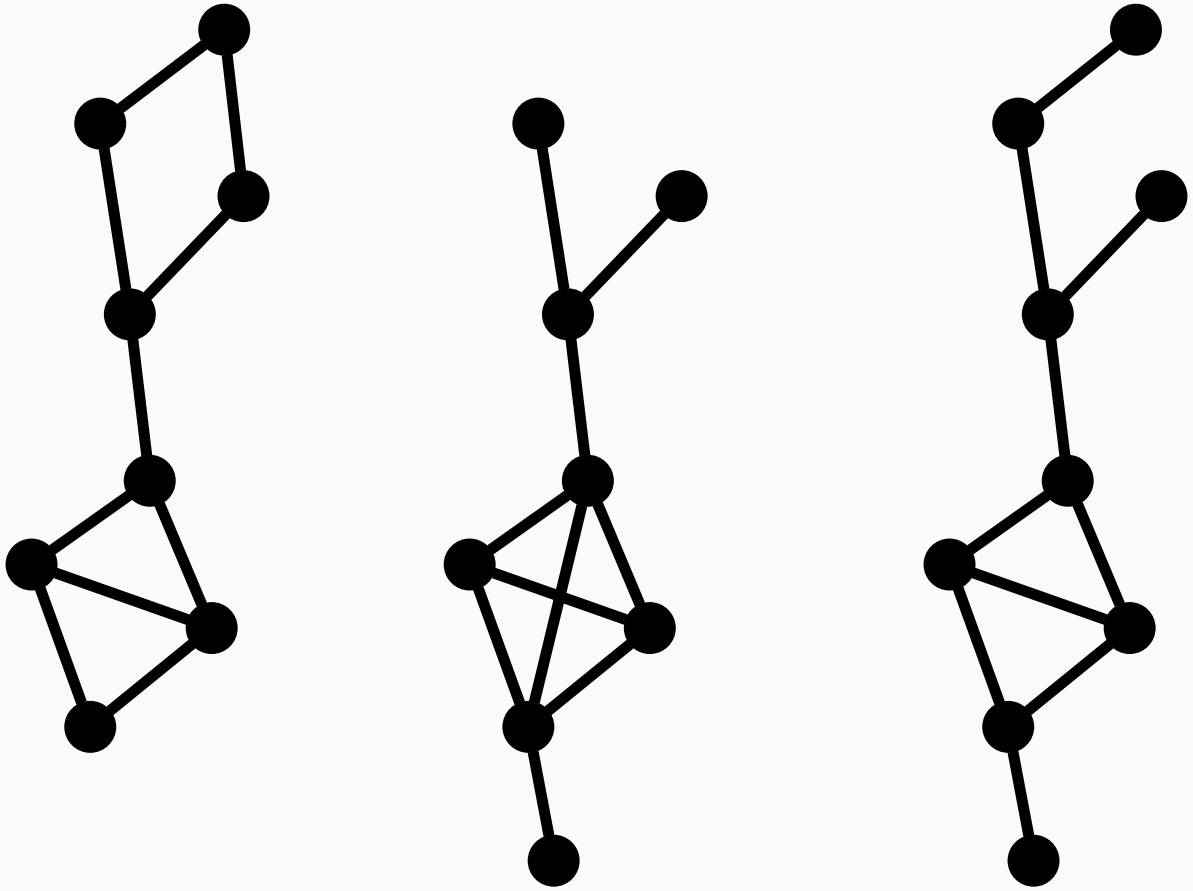
- Identify common core among species.
- Identify functional orthologs.
- 100s of 1000s of edges.
- Topology similar to many types of real-world networks.



# Problem definition

**Multiple maximum common edge subgraph problem**

Given a set of $n$ graphs $\mathbf{X} = \{G_1, G_2, \ldots, G_n\}$ find the largest graph $H$ isomorphic to a subgraph of each graph in $\mathbf{X}$.



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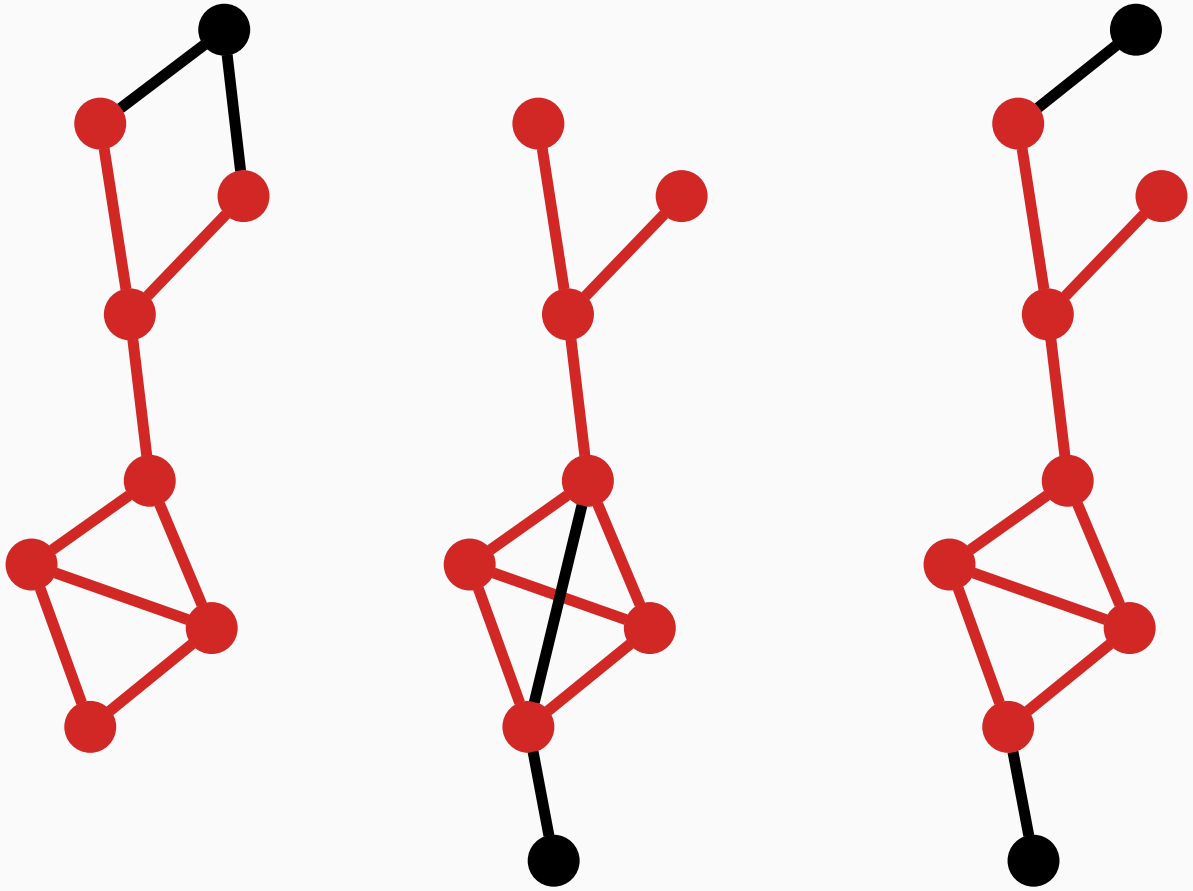

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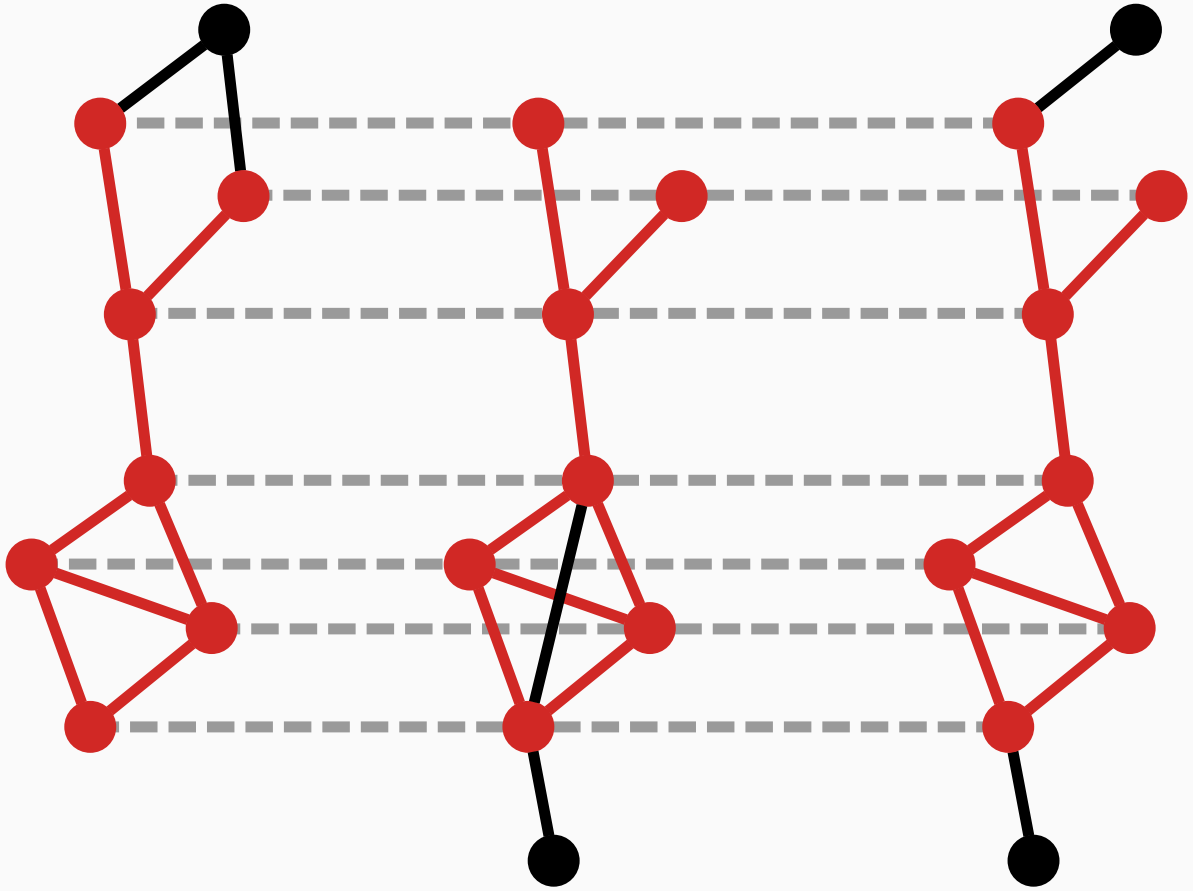


# Related work

*Exact* MCES a well-studied problem with large number of existing algorithms.

Previous work mainly focused on finding common substructures in chemical structures.

No existing heuristic algorithms for multi-MCES of large graphs.

Existing network alignment tools not suited for MCES detection.

# Algorithm outline

Iterated local search algorithm:

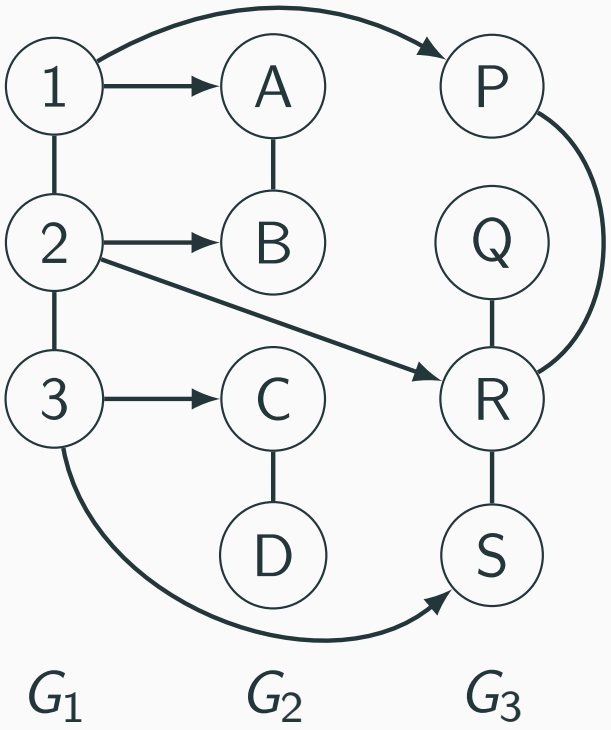
- Repeated greedy local search
- Perturbate solution using pheromones.
- Simulated annealing acceptance function

```
RandomAlignment()
while Time() < MaxTime do
    Perturbate()
    LocalSearch()
    if accepted then
        UpdateSolution()
        UpdatePheromones()
    end
end
```

# Alignment definition

We define an *alignment* as a set of injective functions

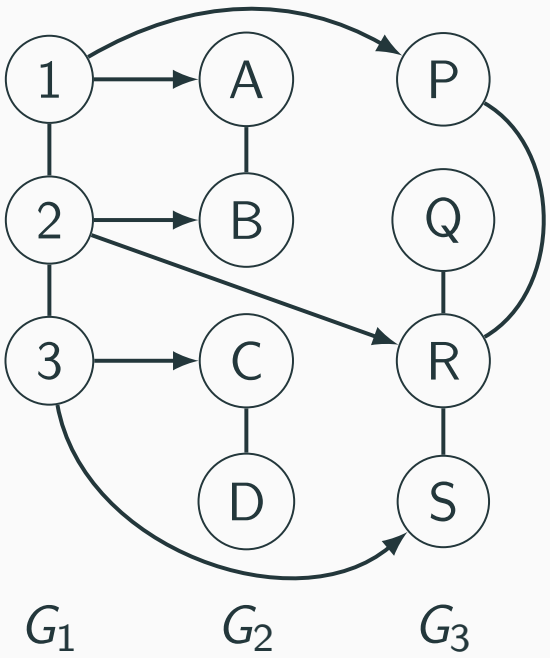
$$A = \{f_i : V_1 \to V_i \mid 1 \leq i \leq n\}, \quad f_1(x) = x.$$

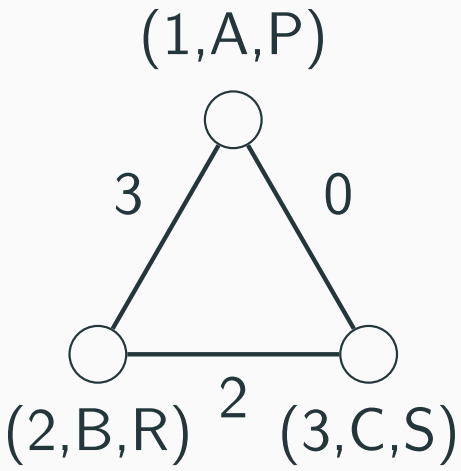


# Alignment definition

Any (valid) alignment corresponds to a candidate solution.

$$\text{Edge count: } C(u,v) = \sum_{i=1}^{n} \begin{cases} 0 & \text{if } (f_i(u), f_i(v)) \in E_i \\ 1 & \text{otherwise.} \end{cases}$$



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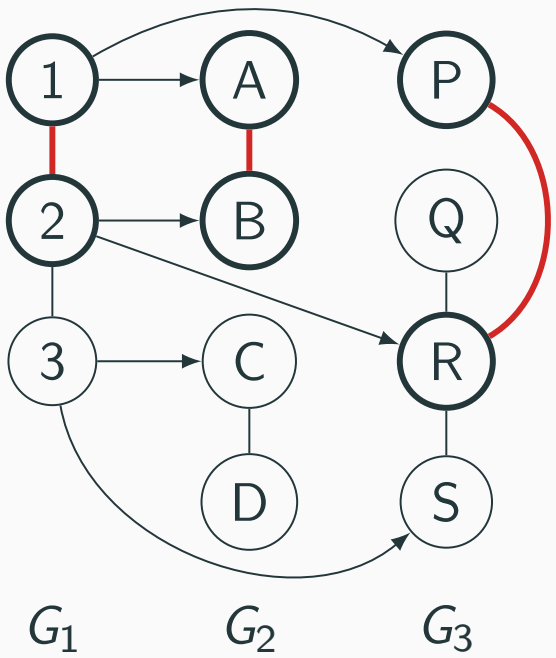


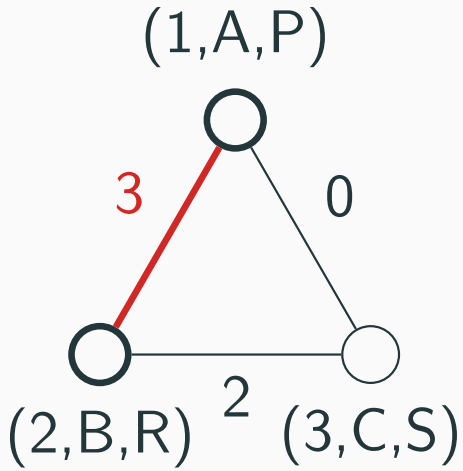

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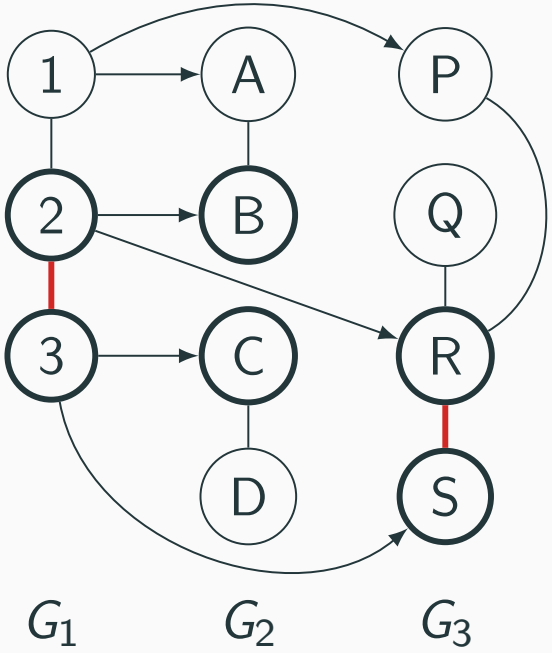


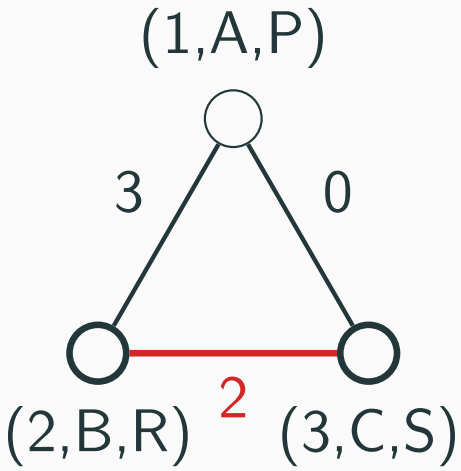

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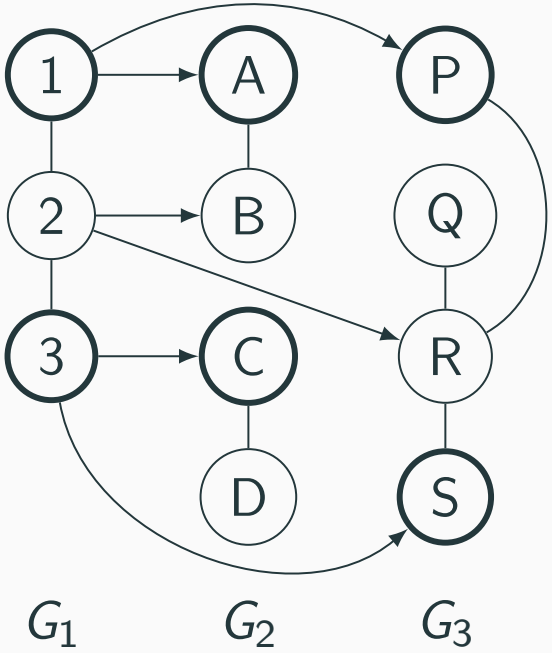


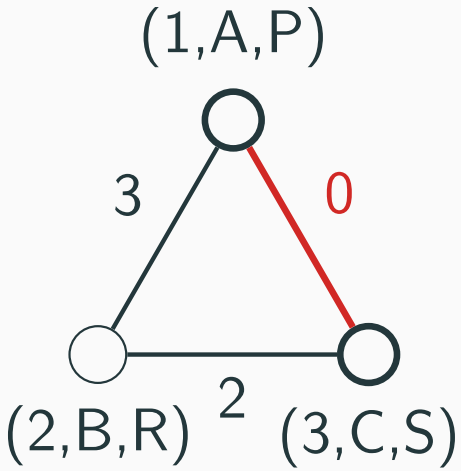

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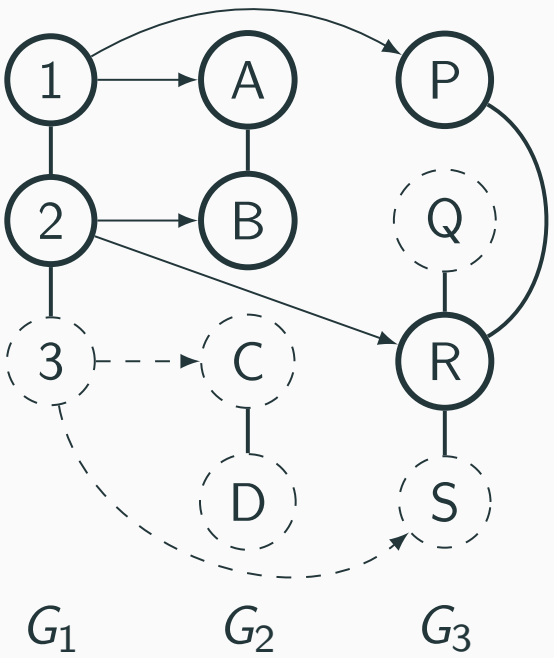
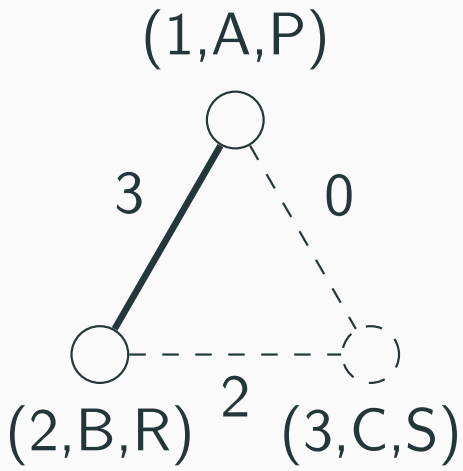


# Local search heuristic

Swap alignment of vertex pairs in same graph.

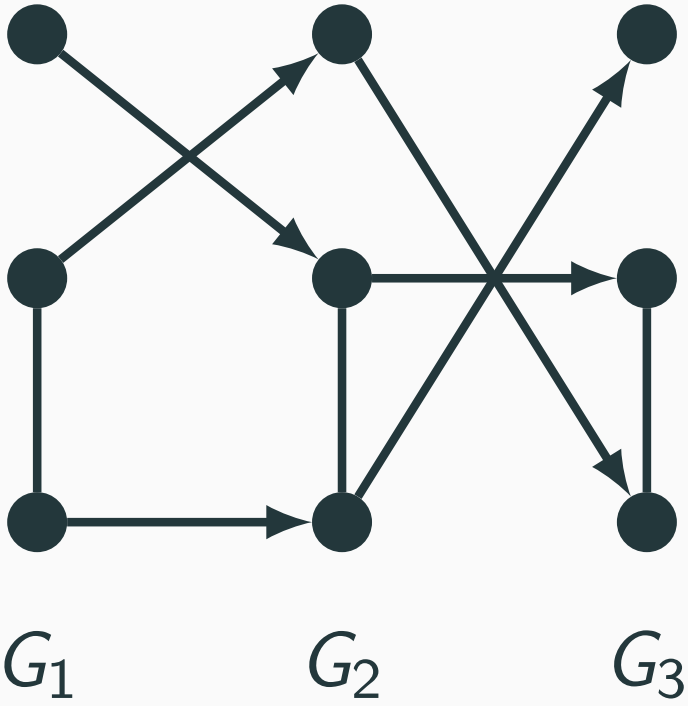
Optimizes following function:

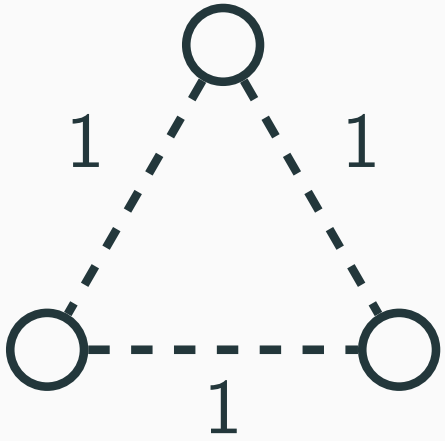
$$g(A) = \sum_{u,v \in V(G_1)} C(u,v)^2$$

Fitness function $F(A)$ not usable. (number of edges)

Local search inspects vertex pairs graph-wise. Repeats until no improving move is found.

# Local search heuristic

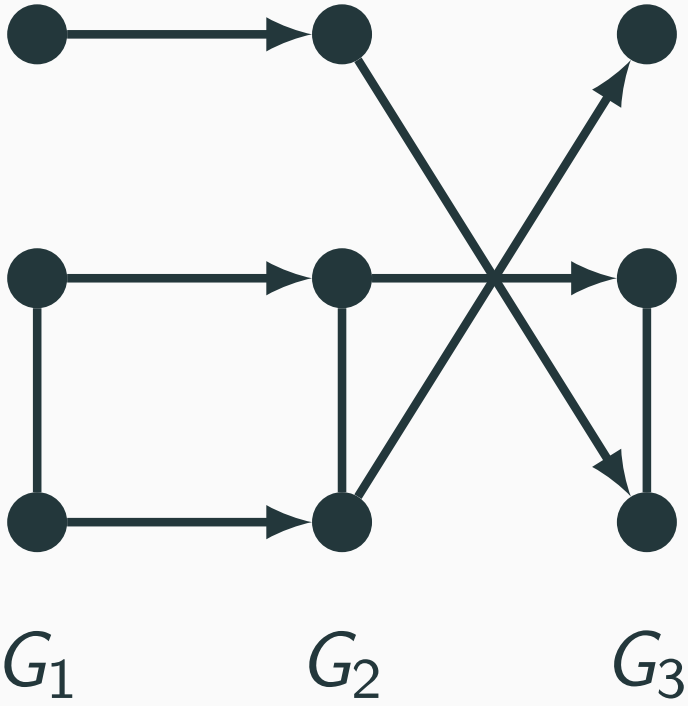


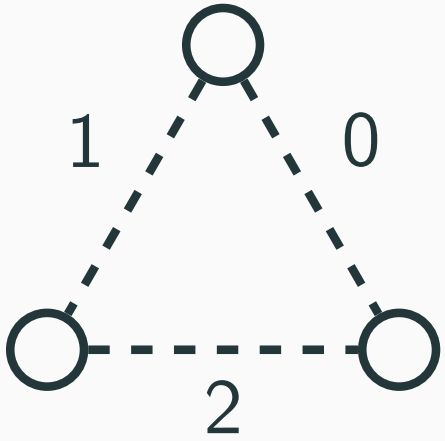




$$g(A) = 1^2 + 1^2 + 1^2 = 3$$

$$F(A) = 0$$

# Local search heuristic

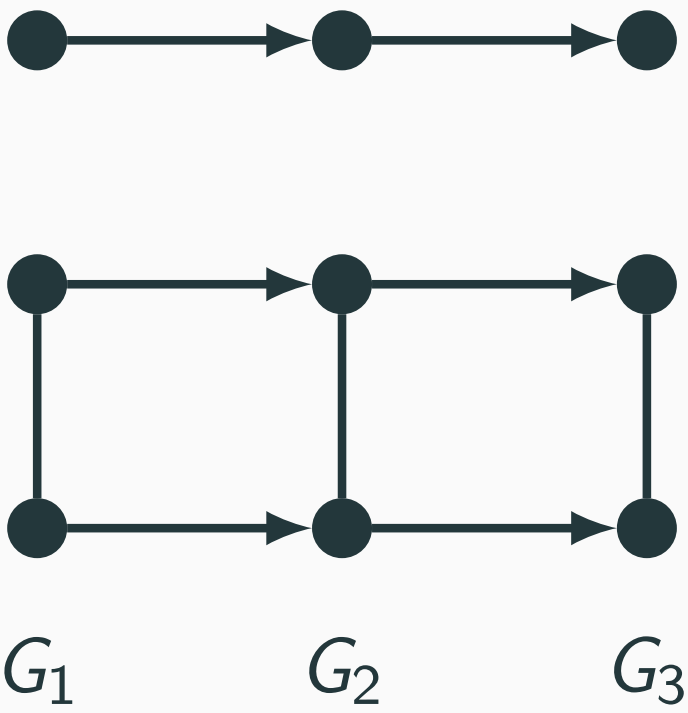


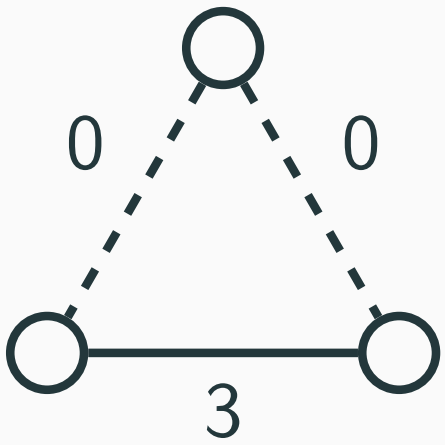




$$g(A) = 1^2 + 0^2 + 2^2 = 5$$

$$F(A) = 0$$

# Local search heuristic




$$g(A) = 0^2 + 0^2 + 3^2 = 9$$

$$F(A) = 1$$

# Local search strategies

**First improvement:**

Swap two vertices $u$ and $v$ immediately when

$$\Delta g(u, v, A) > 0.$$

**Best improvement:**

Swap only the best pair:

$$u, v = \underset{u,v \in V(G_k)}{\arg\max}\ \Delta g(u, v, A).$$

**Vertex-best improvement:**

For each vertex $u \in V(G_k)$ swap with

$$v = \underset{v \in V(G_k)}{\arg\max}\, \Delta g(u, v, A)$$

# Perturbation

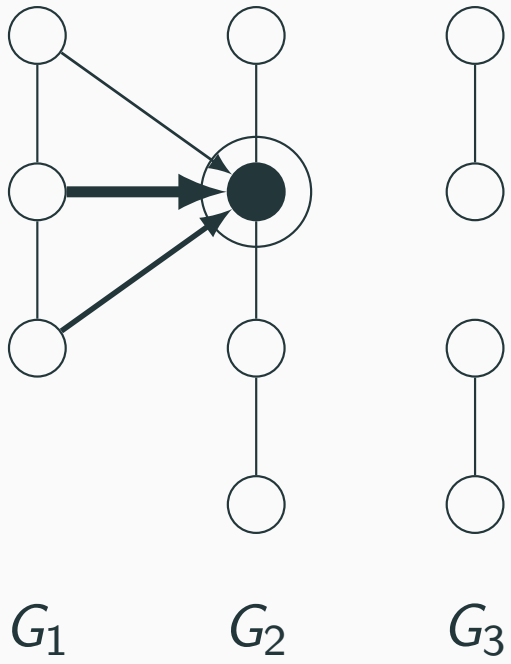
Perturbation based on *MAX-MIN Ant System* (Stützle *et al.*).

Generate new alignment using pheromone matrices $\{\tau_{ij}^2, \tau_{ij}^3, \cdots \tau_{ij}^n\}$.

Align $v_i \in V(G_1)$ to $v_j \in V(G_k)$ with probability $\tau_{ij}^k$.

Pheromones evaporate over time.

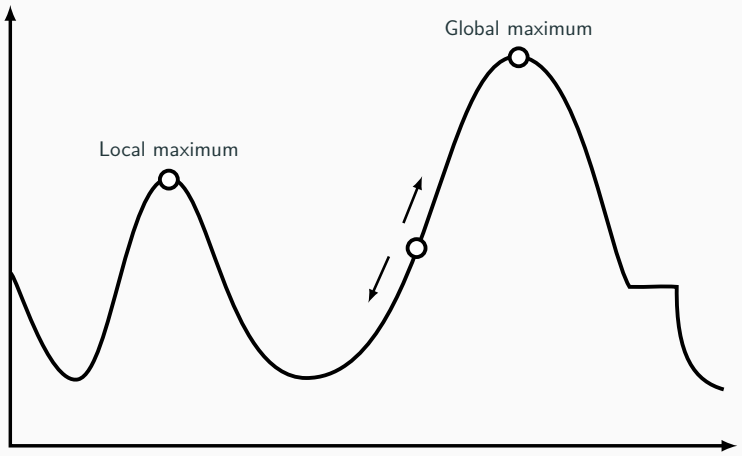
Add pheromones when new solution is accepted.



# Simulated annealing

Introduce simulated annealing acceptance function to:

1. Better explore solution space.
2. Help escape local maxima.




Moves from solution $A$ to new solution $A'$ with probability:

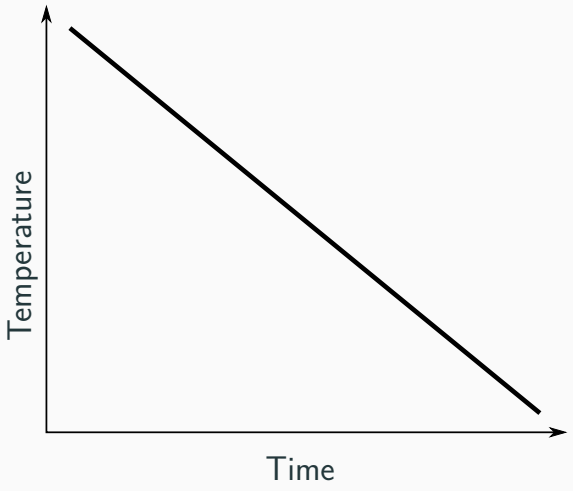
$$P(A, A', T) = \begin{cases} 1 & \text{if } F(A') \geq F(A), \\ e^{(F(A')-F(A))/T} & \text{if } F(A') < F(A). \end{cases}$$

# Simulated annealing cooling schedules

**Linear cooling:**

$$T(t) = \left(1 - \frac{1}{\text{MaxTime}}\right) \cdot T_0$$

$T_0$ – start temperature.

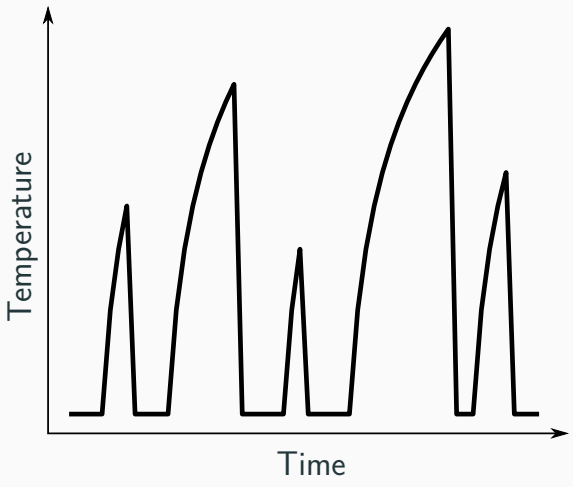



**Adaptive simulated annealing:**

$$T(t) = \lambda \ln(1 + r_i) + T_{min},$$

$T_{min}$ – minimum temperature

$r_i$ – number of consecutive non-improving moves.



# Evaluation

Internal evaluation of different strategies.

Data set:

1. Set of scale-free random graphs.
2. Set of protein-protein interaction networks from 4 species from BioGRID database.

Fixed time limit for all experiments.

| Instance | Vertices | Edges |
|---|---|---|
| Synthetic | | |
| sf1000-1 | 1000 | 1688 |
| sf1000-2 | 1000 | 1681 |
| sf1000-3 | 1000 | 1671 |
| sf1000-4 | 1000 | 1674 |
| sf5000-1 | 5000 | 9434 |
| sf5000-2 | 5000 | 9486 |
| sf5000-3 | 5000 | 9288 |
| sf5000-4 | 5000 | 8921 |
| sf15000-1 | 15 000 | 28 048 |
| sf15000-2 | 15 000 | 27 431 |
| sf15000-3 | 15 000 | 27 633 |
| sf15000-4 | 15 000 | 28 479 |
| Real-world | | |
| rat | 3352 | 4558 |
| mouse | 8706 | 18 814 |
| yeast | 6514 | 227 893 |
| human | 19 803 | 203 104 |

# Evaluation

| Instance | Time | FI-Lin | BI-Lin | VB-Lin | FI-Adp | BI-Adp | VB-Adp |
|---|---|---|---|---|---|---|---|
| sf1000-[1,2] | 10m | 1165 | 1119 | 1164 | 1133 | 1117 | **1171** |
| sf1000-[1,2,3] | 30m | 872 | 777 | **905** | 849 | 812 | 893 |
| sf1000-[1,2,3,4] | 2h | 669 | 610 | 689 | 678 | 544 | **698** |
| sf5000-[1,2] | 30m | 5536 | 5396 | **5568** | 5526 | 5447 | 5555 |
| sf5000-[1,2,3] | 1h | **3718** | 3634 | 3651 | 3608 | 3485 | 3236 |
| sf5000-[1,2,3,4] | 4h | 2463 | 2366 | 2489 | 2703 | 2340 | **2982** |
| sf15000-[1,2] | 1h | 14856 | — | **15757** | 14900 | — | 15741 |
| sf15000-[1,2,3] | 2h | 8662 | — | 5873 | **8918** | — | 5417 |
| sf15000-[1,2,3,4] | 5h | 4685 | — | **7064** | 3947 | — | 6041 |
| sf1000-[1,1] | 10m | **1688** | 1577 | **1688** | 1687 | 1686 | 1687 |
| sf1000-[1,1,1] | 30m | **1282** | 784 | 823 | 863 | 760 | 833 |
| sf1000-[1,1,1,1] | 2h | 723 | 392 | 755 | **762** | 505 | 753 |
| sf5000-[1,1] | 30m | 9280 | 9233 | 6072 | **9293** | 9228 | 9272 |
| sf5000-[1,1,1] | 1h | 3837 | 3632 | 3607 | **3853** | 3304 | 3554 |
| sf5000-[1,1,1,1] | 4h | 2975 | 2518 | **3200** | 2495 | 2511 | 2841 |
| sf15000-[1,1] | 1h | 15423 | — | 16717 | 15578 | — | **20168** |
| sf15000-[1,1,1] | 2h | 8612 | — | 8579 | 8556 | — | **9386** |
| sf15000-[1,1,1,1] | 5h | 6571 | — | **6855** | 5065 | — | 4519 |
| rat, mouse | 1h | 2333 | **2726** | 2637 | 2469 | 2512 | 2581 |
| rat, mouse, yeast | 2h | 1043 | 1043 | **1597** | 1148 | 1272 | 1524 |
| rat, mouse, yeast, human | 5h | 483 | 820 | 669 | 599 | 702 | **988** |

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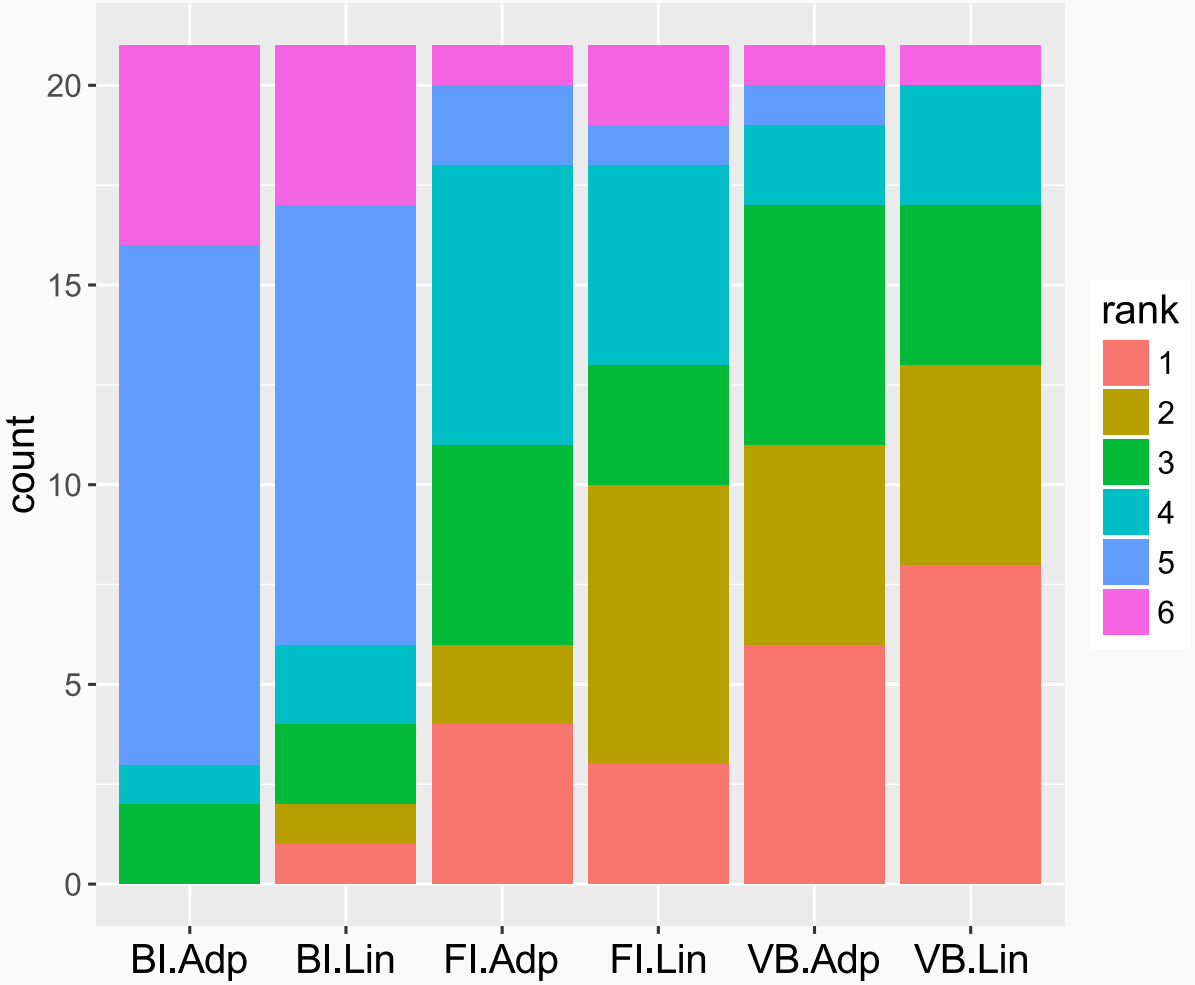
# Evaluation




**Avg. rank:**

| BI-Adp | BI-Lin | FI-Adp | FI-Lin | VB-Adp | VB-Lin |
|---|---|---|---|---|---|
| 5.00 | 4.57 | 3.19 | 3.00 | **2.52** | **2.29** |

# Contribution of pheromone-based perturbation

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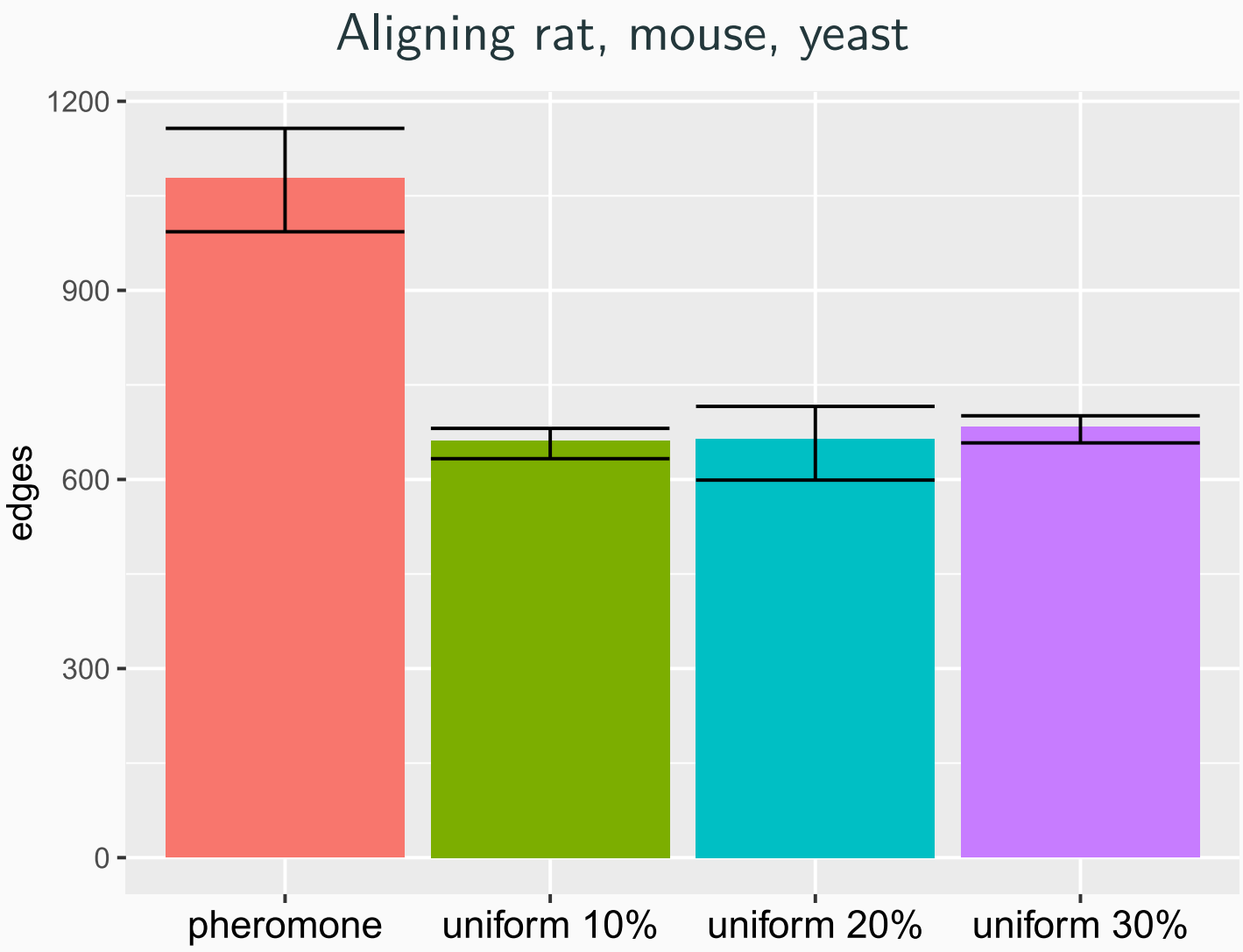

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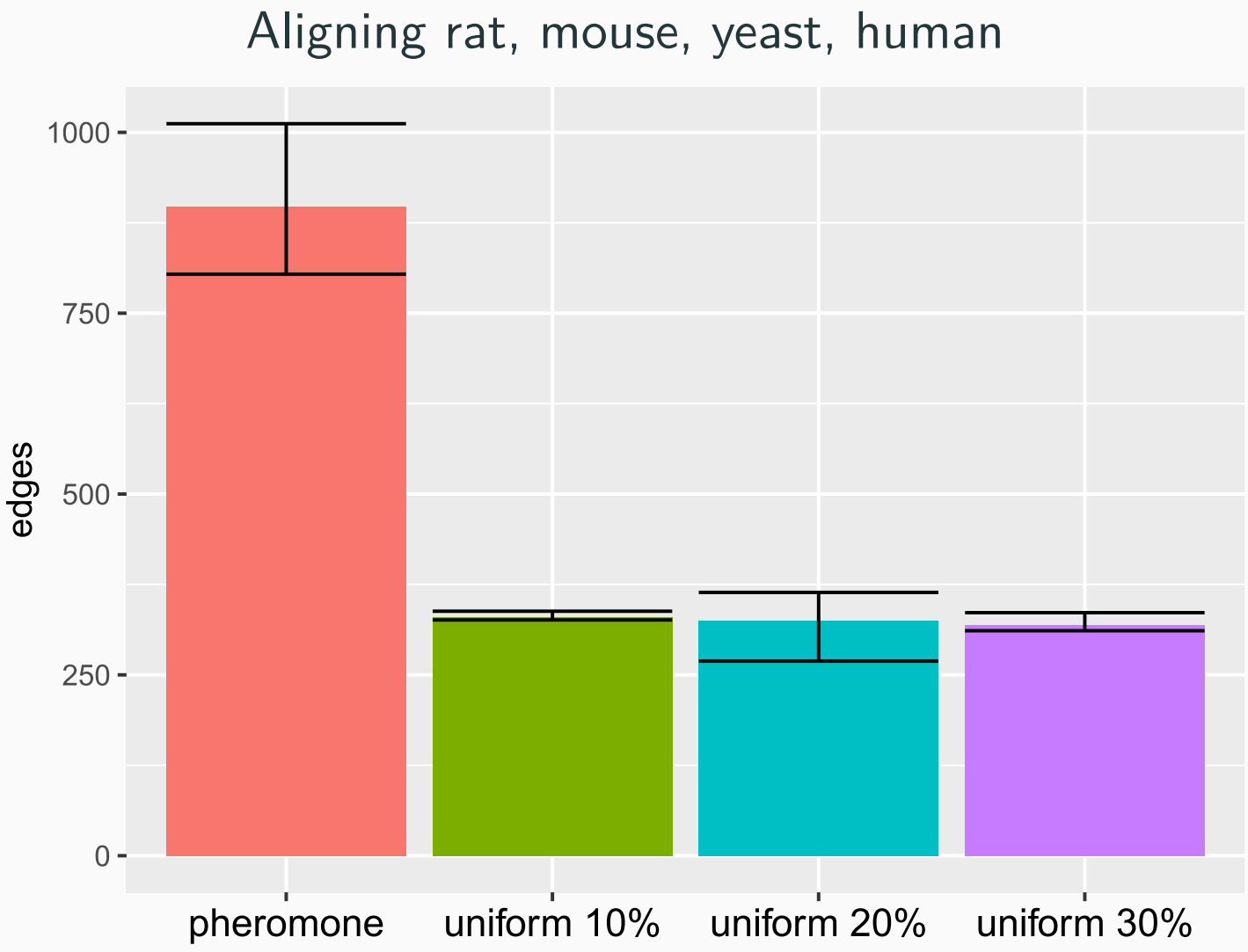
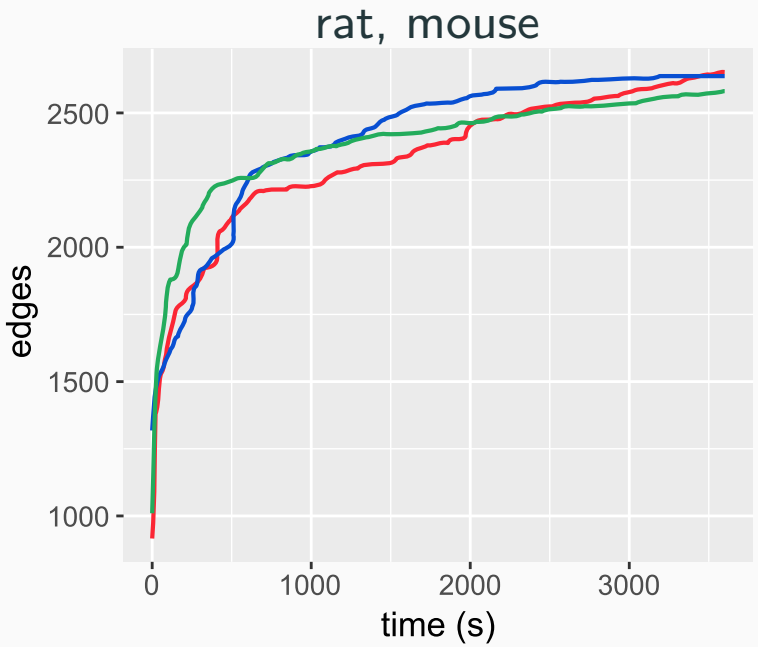
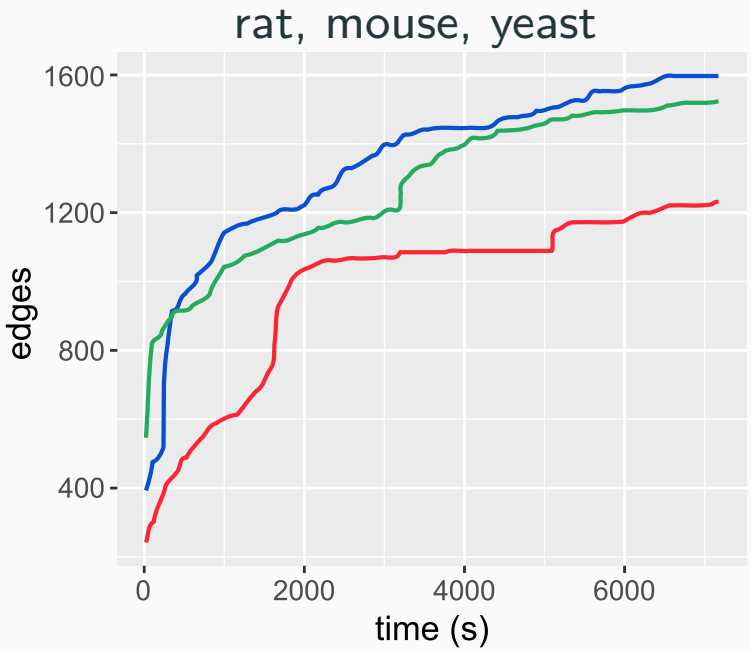


# Comparison of annealing schedules



# Conclusions

- Heuristic algorithm for multiple MCES detection.
- Vertex-best local search strategy superior to first improvement and best improvement.
- Best improvement strategy failed to produce results in large instances.
- Pheromone-based perturbation significant improvement with larger problem instances.
- Annealing acceptance criterion provided small benefit to algorithm.

# Thank you!

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**Source code and test data:**

http://sailmcs.compbio.sdu.dk